Assignment Rectangles and Sampling

Image Analysis Assignment 2 -Group 20

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In this practical, we investigated the effects of discretization and noise on the measurement of rectangular objects in grayscale images. Three series of images (A: clean, B and C: with increasing noise) were analyzed. We performed threshold-based segmentation, measured area and perimeter, evaluated discretization error via coefficient of variation, applied noise suppression filters (Median, Gaussian and Kuwahara with varying kernel sizes), computed signal-to-noise ratios (SNR), and related SNR to measurement accuracy.

Part 2.1

1. Histograms

Figure 1 shows the histograms of rect1a to rect4a. All images show narrow, distinct intensity peaks due to the limited intensity distribution of the rectangular objects. A critical observation is that the foreground objects have a lower intensity value (approximately 62), meaning they are darker than the background (approximately 192).

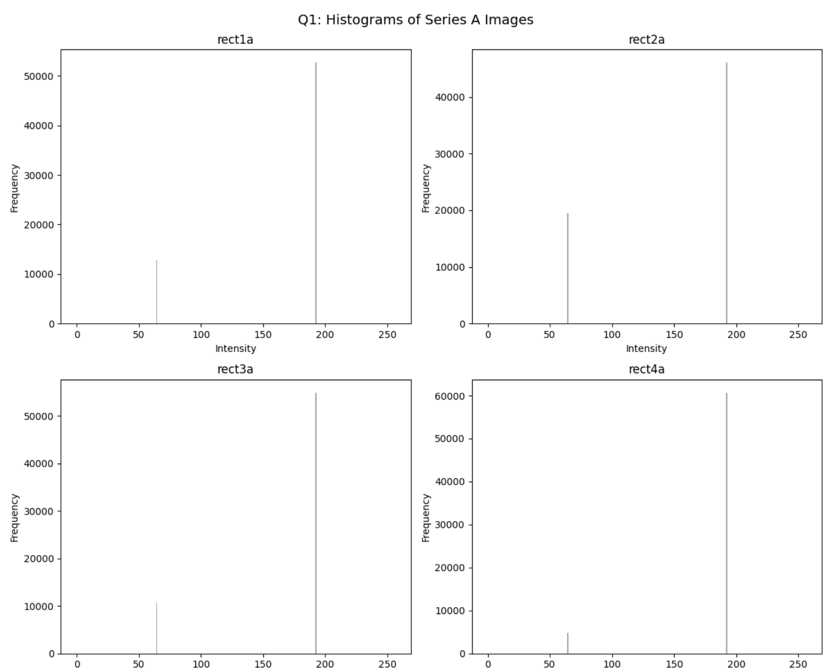

Figure 1: Histograms of series A images.

2. Area Measurements

The area of each object was measured by first thresholding the images using the Isodata method, followed by connected component labeling. The solid area of each labeled object was then computed. Table 1 presents the individual object measurements, while Table 2 summarizes the mean and standard deviation per image.

Table 1: Measured area and perimeter for each object in all images

Image	Object ID	Area	Perimeter
rect1a	1	12848	457.50
rect2a	1	3195	233.73
rect2a	2	3285	230.14
rect2a	3	3265	234.68
rect2a	4	3248	235.98
rect2a	5	3230	233.27
rect2a	6	3262	235.07
rect3a	1	1320	143.90
rect3a	2	1328	148.77
rect3a	3	1335	148.86
rect3a	4	1304	146.93
rect3a	5	1340	147.01
rect3a	6	1309	147.54
rect3a	7	1297	147.16
rect3a	8	1360	150.47
rect4a	1	475	92.43
rect4a	2	480	90.98
rect4a	3	480	90.98
rect4a	4	480	92.92
rect4a	5	486	93.35
rect4a	6	473	92.05
rect4a	7	468	92.67
rect4a	8	499	93.54
rect4a	9	484	94.08
rect4a	10	505	95.23

3. Perimeter Measurements

Perimeter measurements were computed using the same segmentation pipeline, with perimeter defined as the boundary length of each labeled object. The results are included in Tables 1 and 2.

Observations from Table 2: As expected, rect1a contains only a single large object, resulting in zero standard deviation for both area and perimeter. As the object size decreases from rect1a to rect4a, both the mean area and mean perimeter decrease, while the standard deviation relative to the mean increases, indicating the onset of discretization effects.

Table 2: Summary statistics (mean and standard deviation) of area and perimeter per image

Image	Object Count	Mean Area	Std Area	Mean Perimeter	Std Perimeter
rect1a	1	12848.00	0.00	457.50	0.00
rect2a	6	3247.50	28.83	233.81	1.86
rect3a	8	1324.13	19.55	147.58	1.80
rect4a	10	483.00	10.80	92.82	1.26

Part 2.2: Relative Discretization Error

4. Area vs Coefficient of Variation

Table 3 presents the data used to evaluate the relative discretization error for area measurements. The coefficient of variation is defined as $cv = \sigma/\mu$. Figure 2 plots the square root of the mean area against the CV.

Image	Mean Area (μ)	Std (σ)	CV (σ/μ)	$\sqrt{\mu}$
rect1a	12848.00	0.00	0.0000	113.35
rect2a	3247.50	28.83	0.0089	56.99
rect3a	1324.13	19.55	0.0148	36.39
rect4a	483.00	10.80	0.0224	21.98

Table 3: Relative discretization error data for area measurements.

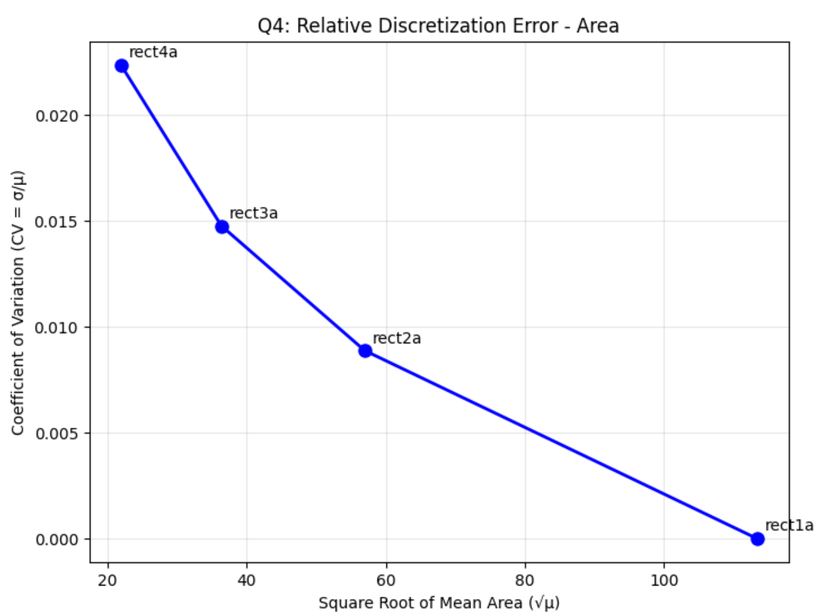

Figure 2: Relative discretization error for area: Coefficient of variation versus square root of mean area.

As shown in Figure 2, the coefficient of variation increases as object size decreases. The largest object (rect1a) has zero variance, as expected for a single, well-resolved shape. Smaller objects exhibit greater relative variability due to discretization effects: fewer pixels represent their boundaries, making area measurements more sensitive to pixel-level quantization.

5. Perimeter vs Coefficient of Variation

Table 4 and Figure 3 show the analogous relationship for perimeter measurements.

Image	Mean Perimeter (μ)	Std (σ)	CV (σ/μ)	$\sqrt{\mu}$
rect1a	457.50	0.00	0.0000	21.39
rect2a	233.81	1.86	0.0080	15.29
rect3a	147.58	1.80	0.0122	12.15
rect4a	92.82	1.26	0.0135	9.63

Table 4: Relative discretization error data for perimeter measurements.

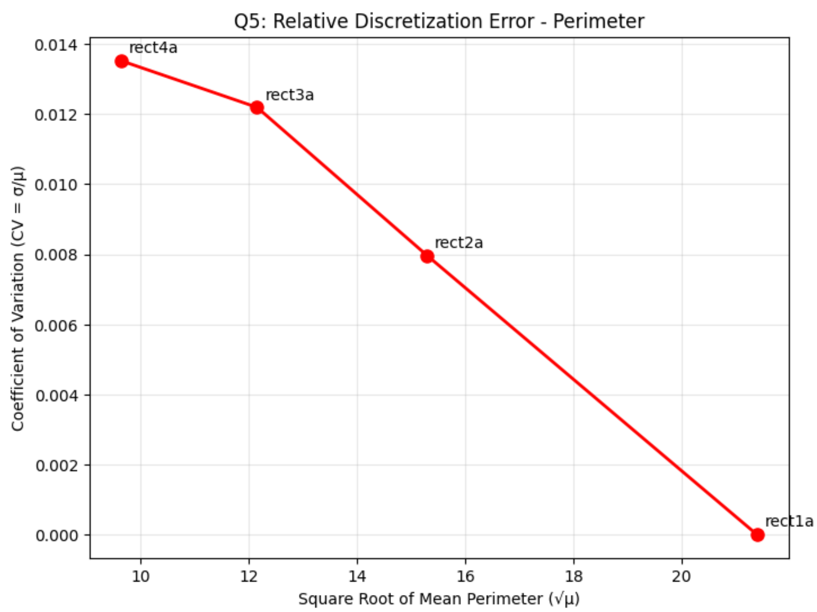

Figure 3: Relative discretization error for perimeter: Coefficient of variation versus square root of mean perimeter.

Perimeter measurements show the same inverse trend, with CV decreasing as object size increases. However, the curve is slightly steeper, indicating that perimeter is somewhat more sensitive to discretization effects than area for very small objects.

6. Comparison of Area and Perimeter Errors

As shown in Tables 3 and 4, the coefficient of variation (CV) for area measurements is consistently higher than for perimeter measurements across all object sizes. For rect4a, area CV = 0.0224 versus perimeter CV = 0.0135; for rect3a, 0.0148 versus 0.0122; for rect2a, 0.0089 versus 0.0080. This indicates that discretization errors have a greater relative impact on area than on perimeter.

The difference between these plot include that the area scales quadratically with object dimensions, making it more sensitive to pixel-level boundary shifts than perimeter, which scales linearly. A single-pixel error along the boundary changes the area by approximately the boundary length (in pixels), while the perimeter changes by only a few pixel units. Therefore, the relative discretization error (CV) is larger for area measurements.

Part 2.3: Noise Suppression

7. Histograms for Series B and C

Figure 4 displays histograms for the noisy series. Series B (moderate noise) shows broadened intensity distributions, while Series C (high noise) exhibits even wider spread.

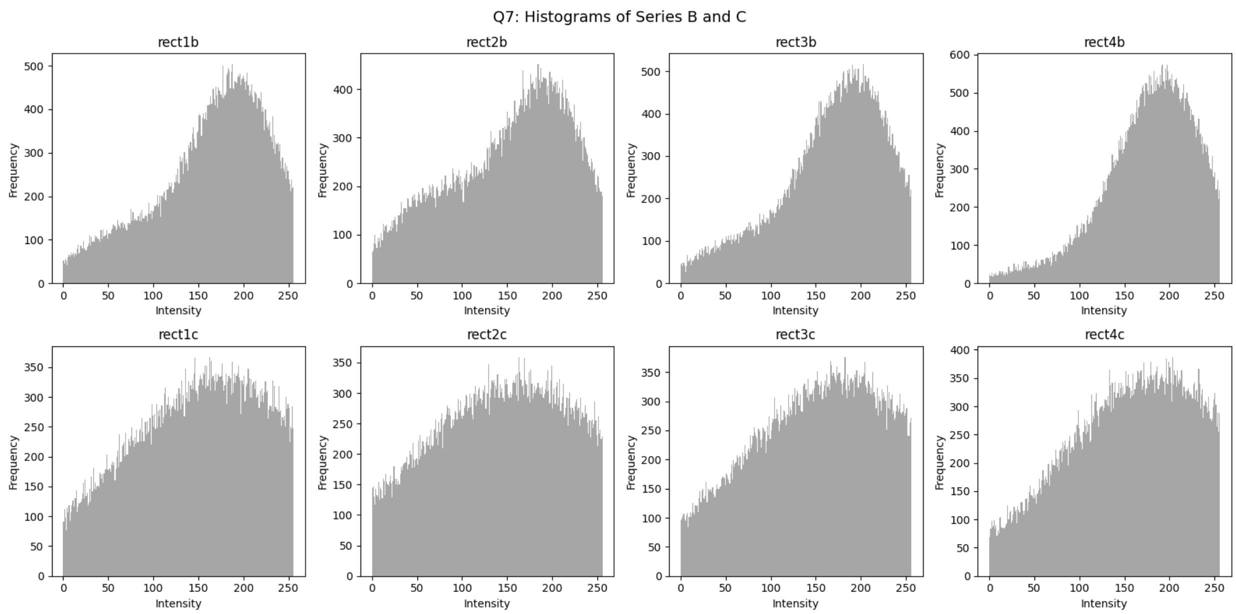

Figure 4: Histograms of Series B (top row) and Series C (bottom row) images. Note the increasing spread and loss of distinct peaks with higher noise.

8. Noise Suppression Filter Selection

We evaluated three filters with two parameter settings each (Figure 5):

- **Median filter**: kernel sizes 3 and 7
- **Gaussian filter**: $\sigma = 3$ $\sigma = 5$
- **Kuwahara filter**: kernel sizes 3 and 5

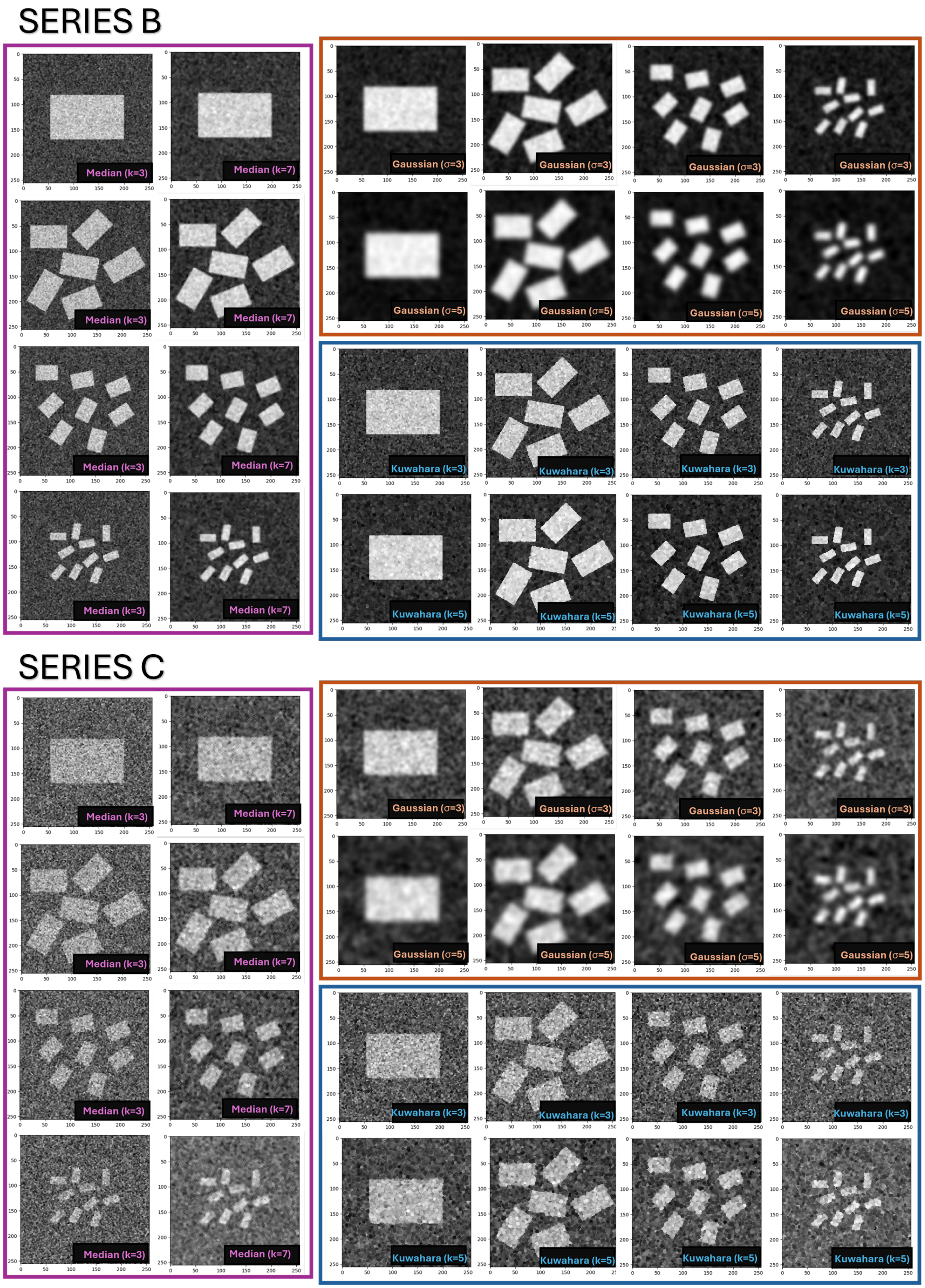

Figure 5: Filters and parameters of Series B and C images (note image were inverted for easier visualization). On the left column in purple border is the Median filter using kernels 3 and 7. The top right row in orange border in the Gaussian filter using sigmas 3 and 5. The bottom right row in blue border in the Kuwahara filter using kernels of 3 and 5.

9. Repeat measurements after filtering

Tables 5 and 6 present the area and perimeter measurements after applying each filter. Figures 6 and 7 plot the relative discretization errors for the filtered results.

Image	Filter	Mean Area	Std Area	Mean Perim	Std Perim	No. Objects
rect1b	Gaussian_k3	12848.00	0.00	453.31	0.00	1
rect1b	Gaussian_k5	12946.00	0.00	450.59	0.00	1
rect1b	Kuwahara_k3	1075.50	3552.26	45.07	126.92	12
rect1b	Kuwahara_k5	12870.00	0.00	465.28	0.00	1
rect1b	Median_k3	231.14	1702.47	12.40	62.26	56
rect1b	Median_k7	12870.00	0.00	468.57	0.00	1
rect2b	Gaussian_k3	3260.17	30.02	226.99	1.57	6
rect2b	Gaussian_k5	3308.00	36.86	223.99	2.38	6
rect2b	Kuwahara_k3	1216.50	1565.00	94.63	113.63	16
rect2b	Kuwahara_k5	3240.50	19.26	235.26	1.92	6
rect2b	Median_k3	296.77	933.50	25.71	68.56	66
rect2b	Median_k7	3246.83	25.23	233.12	1.51	6
rect3b	Gaussian_k3	1350.25	21.22	141.54	1.17	8
rect3b	Gaussian_k5	1403.00	20.21	139.68	1.23	8
rect3b	Kuwahara_k3	1054.30	526.39	121.67	57.93	10
rect3b	Kuwahara_k5	1313.75	18.52	146.25	1.48	8
rect3b	Median_k3	152.56	420.40	21.11	47.58	70
rect3b	Median_k7	1316.00	28.47	144.56	1.36	8
rect4b	Gaussian_k3	536.90	14.07	90.41	1.02	10
rect4b	Gaussian_k5	828.62	303.53	116.34	35.10	8
rect4b	Kuwahara_k3	320.33	224.59	65.38	42.26	15
rect4b	Kuwahara_k5	474.50	9.90	91.42	1.72	10
rect4b	Median_k3	43.20	135.43	11.91	25.54	116
rect4b	Median_k7	481.30	13.54	90.05	1.63	10

Table 5: Results for Series B

Image	Filter	Mean Area	Std Area	Mean Perim	Std Perim	No. Objects
rect1c	Gaussian_k3	6425.00	6401.00	238.99	226.20	2
rect1c	Gaussian_k5	12849.00	0.00	453.11	0.00	1
rect1c	Kuwahara_k3	21.18	406.11	11.96	27.43	1052
rect1c	Kuwahara_k5	58.59	804.69	12.11	37.80	256
rect1c	Median_k3	19.81	374.17	12.77	30.22	1272
rect1c	Median_k7	55.78	809.63	9.51	35.45	254
rect2c	Gaussian_k3	3271.17	29.56	230.41	1.67	6
rect2c	Gaussian_k5	3311.17	41.04	224.22	2.23	6
rect2c	Kuwahara_k3	32.79	294.22	12.56	34.15	781
rect2c	Kuwahara_k5	117.06	588.60	19.29	51.33	182
rect2c	Median_k3	25.15	250.73	11.96	33.37	1115
rect2c	Median_k7	104.48	565.63	15.35	48.21	197
rect3c	Gaussian_k3	1350.38	32.27	145.09	2.31	8
rect3c	Gaussian_k5	1408.88	42.27	141.85	2.45	8
rect3c	Kuwahara_k3	21.64	134.20	14.45	30.62	1012
rect3c	Kuwahara_k5	43.69	217.34	14.92	30.30	310
rect3c	Median_k3	20.67	125.47	14.98	31.45	1202
rect3c	Median_k7	38.05	208.70	12.50	27.40	336
rect4c	Gaussian_k3	445.17	196.93	78.67	31.71	12
rect4c	Gaussian_k5	943.00	516.15	135.09	64.74	7
rect4c	Kuwahara_k3	24.60	124.35	20.45	50.82	948
rect4c	Kuwahara_k5	24.75	104.27	17.79	31.83	598
rect4c	Median_k3	33.86	225.96	25.74	94.90	856
rect4c	Median_k7	26.85	162.37	17.89	48.33	651

Table 6: Results for Series C

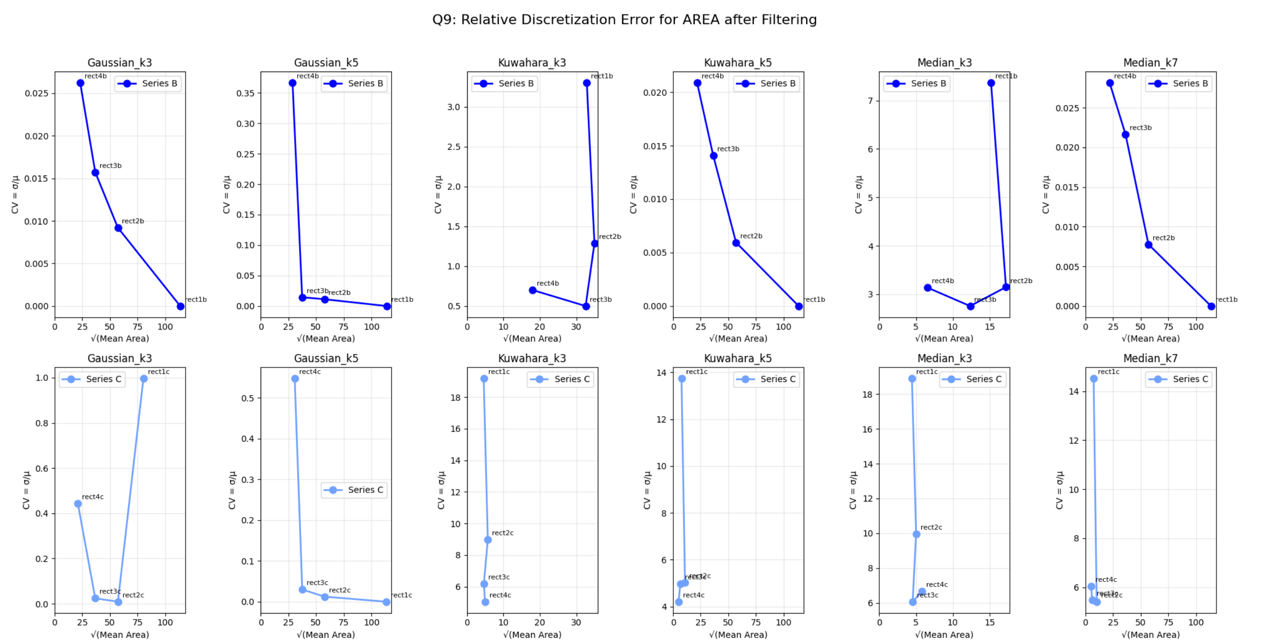

Figure 6: Relative discretization error for area measurements after filtering (Series B and C).

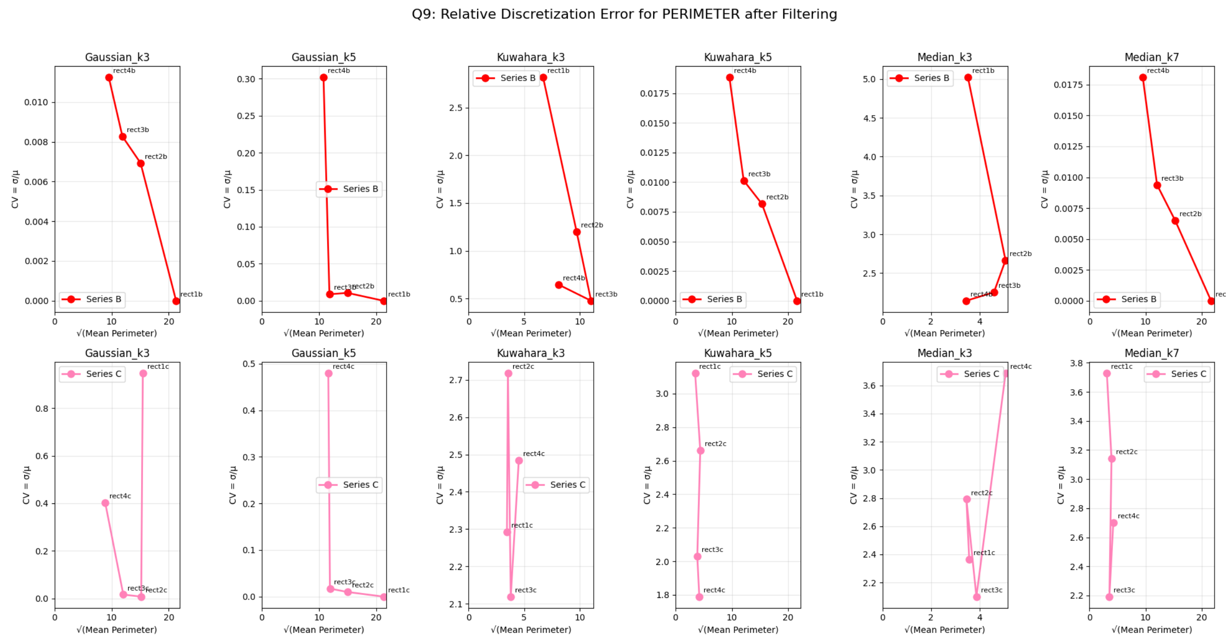

Figure 7: Relative discretization error for perimeter measurements after filtering (Series B and C).

10. Influence of Noise on Measurements

Table 6 compared to Tables 5 and 1 shows that excessive amounts of noise can have disastrous effects for basic measurements such as average area, average perimeter and the number of observed objects in the image, in particular when the noise amplitude exceeds the intensity threshold separating foreground from background. Moderate noise (Series B) causes a slight inflation of measured area and perimeter as noisy boundary pixels are misclassified as foreground, while severe noise (Series C) causes catastrophic over-segmentation, producing hundreds of spurious objects with near-zero area and driving mean measurements far from the true values.

This threshold is more quickly exceeded for smaller foreground objects such as rect3c and rect4c, where the true foreground signal occupies fewer pixels and the same absolute noise amplitude represents a larger fraction of the object. Large noise could therefore trick the object counter into detecting more elements than truly exist, since a single object split by noise suddenly appears as two. One could think of a large football field versus many Jenga pieces: even in a noisy image, the football field remains a good approximation, yet for the Jenga pieces the noise compounds across each object in the scene, quickly corrupting measurements of area, perimeter and object count, especially when the noise suppression filter is insufficient for the noise level present.

11. Sampling Theorem Interpretation

According to the Whittaker–Nyquist–Shannon sampling theorem [1], a signal must be sampled at least twice the highest frequency component to avoid aliasing. In the spatial domain of a digital image, the pixel grid acts as the sampler and object boundaries represent the highest spatial frequency content.

For large objects such as rect1 and rect2, edges span many pixels and their spatial frequency is well below the Nyquist limit, so measurements are accurate with low discretization error. For small objects such as rect4, edges occupy relatively few pixels and their spatial frequency approaches the Nyquist limit, explaining the higher CV values observed in Part 2.2.

When noise is introduced it adds high-frequency content on top of the object signal. For Series B this noise remains distinguishable from the object signal through filtering, and the CV remains relatively stable across object sizes (see Figure 6), meaning the sampling theorem conditions are effectively restored after filtering. For Series C, the noise amplitude is large enough that it occupies the same spatial frequency band as the object boundaries, making faithful reconstruction impossible for small objects regardless of the filter applied. This explains why the CV varies widely for Series C. Furthermore, aggressive filters such as Gaussian with large σ risk fusing adjacent objects by smoothing the gap between them below the detection threshold, analogous to aliasing where two distinct frequency components become indistinguishable after undersampling. This is consistent with the varying object counts observed in Table 6 even across different Gaussian parameters.

Part 2.4

12. SNR Computation and Filter

Tables 7, 8, and 9 present the SNR values (SNR = μ/σ) for original and filtered images. Figure 8 visualizes the SNR improvement achieved by each filter.

Image	Series A SNR	Series B SNR	Series C SNR
rect1	3.2845	2.6590	2.1626
rect2	2.6312	2.3021	2.0429
rect3	3.6357	2.8273	2.1987
rect4	5.4589	3.4282	2.3291
Mean	3.7526	2.8042	2.1833

Table 7: SNR values for original (unfiltered) images across Series A, B, and C. Note the progressive decrease in SNR from A to C.

Image	Median_k3	Median_k7	Gauss_s3	Gauss_s5	Kuwahara_k3	Kuwahara_k5
rect1b	3.3152	3.5230	3.8546	3.9725	3.5129	3.5129
rect2b	2.7640	2.9389	3.2958	3.5313	2.8627	2.8627
rect3b	3.6833	4.0378	4.6228	5.0892	3.9221	3.9221
rect4b	5.1628	6.0744	7.3287	8.5826	5.8413	5.8413
Mean	3.7313	4.1435	4.7755	5.2939	4.0347	4.0347

Table 8: SNR values for filtered Series B (moderate noise)

Image	Median_k3	Median_k7	Gauss_s3	Gauss_s5	Kuwahara_k3	Kuwahara_k5
rect1c	3.7852	4.9369	6.7427	7.0864	5.2228	5.2228
rect2c	3.3685	4.2151	5.8541	6.3798	4.3924	4.3924
rect3c	3.9536	5.3648	7.8320	8.8767	5.6520	5.6520
rect4c	4.5943	7.0756	11.5442	14.3018	7.8206	7.8206
Mean	3.9254	5.3981	7.9933	9.1612	5.7720	5.7720

Table 9: SNR values for filtered Series C (high noise)

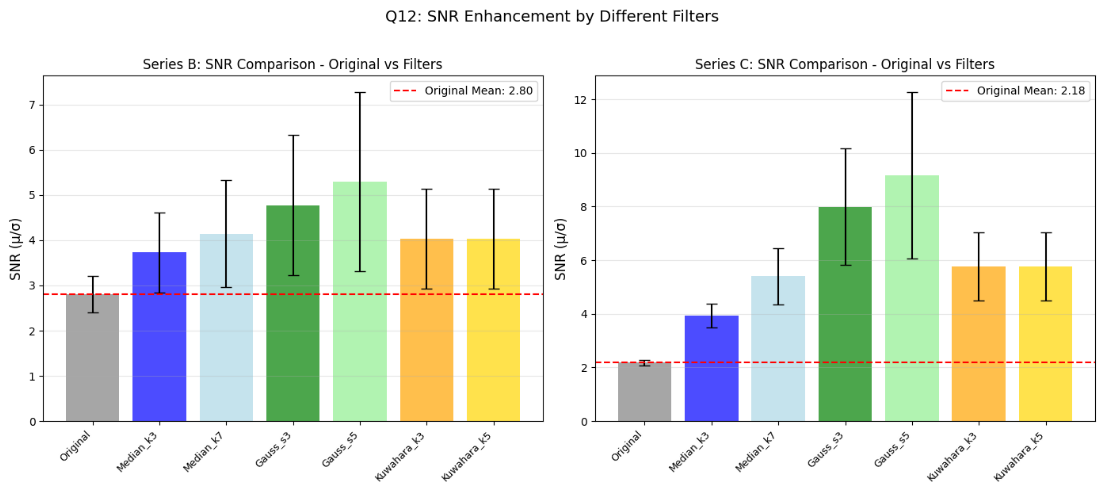

Figure 8: SNR enhancements by different filters. Gaussian filters with larger sigma achieve the highest SNR improvement.

Filter	Mean SNR	Improvement	Improvement (%)
Median_k3	3.7313	0.9272	33.06
Median_k7	4.1435	1.3394	47.76
Gauss_s3	4.7755	1.9713	70.30
Gauss_s5	5.2939	2.4897	88.79
Kuwahara_k3	4.0347	1.2306	43.88
Kuwahara_k5	4.0347	1.2306	43.88

Table 10: SNR improvement for Series B (original mean SNR = 2.8042)

Best filter: Gauss_s5 (Improvement = 2.4897, 88.79%)

Filter	Mean SNR	Improvement	Improvement (%)
Median_k3	3.9254	1.7421	79.79
Median_k7	5.3981	3.2148	147.24
Gauss_s3	7.9933	5.8100	266.11
Gauss_s5	9.1612	6.9779	319.60
Kuwahara_k3	5.7720	3.5886	164.37
Kuwahara_k5	5.7720	3.5886	164.37

Table 11: SNR improvement for Series C (original mean SNR = 2.1833)

Best filter: Gauss_s5 (Improvement = 6.9779, 319.60%)

As shown in Tables 7–9, the SNR decreases progressively from Series A through Series C, confirming that the noise level in Series C is substantially higher than in Series B. For example, the mean SNR drops from 3.75 (Series A) to 2.80 (Series B) to 2.18 (Series C).

After filtering, all filters improve SNR above the original values, but the degree of improvement varies considerably. Gaussian filtering with $\sigma = 5$ achieves the highest SNR improvement in both series: an 88.79% improvement for Series B and a 319.60% improvement for Series C. The Gaussian filter is effective because it applies a smooth, weighted average over a wide neighbourhood, suppressing high-frequency noise, which is precisely what broadens the intensity distributions seen in the Series B and C histograms. The larger $\sigma = 5$ kernel integrates over more pixels, attenuating noise more aggressively than $\sigma = 3$.

The Median k3 and Kuwahara k3 filters perform least well (33% and 44% for Series B respectively). For Series C, small-kernel filters are largely insufficient: with only a 3×3 neighbourhood, the filter cannot average over enough pixels to overcome the heavy noise. The conclusion is that Gaussian with $\sigma = 5$ is the most effective filter for noise suppression in these images, particularly when noise is severe, at the cost of some spatial resolution due to the blurring effect.

13. SNR vs. Measurement Accuracy

Figure 9 presents six plots relating SNR to the logarithm of total measured area and perimeter per image, organized as: original unfiltered images (top row), Series B filtered results (middle row), and Series C filtered results (bottom row).

Original images (top row). In the unfiltered plots, the three series form distinct clusters. Series A (clean) points are spread across a wide SNR range (2.6–5.5) but occupy lower log(area) values, since objects are correctly segmented without spurious detections. Series B and C points cluster at higher log(area/perimeter) values (highlighted by the orange boxes) because noise causes false foreground detections, inflating the total measured area and perimeter beyond the true values. Series C (red square/circle) sits at the lowest SNR values while still showing inflated measurements, confirming that high noise simultaneously degrades signal quality and corrupts segmentation. The anomalous position of rect4a (top-left in both plots) reflects that small objects with few background pixels have a naturally high pixel-intensity SNR, yet their total area is small.

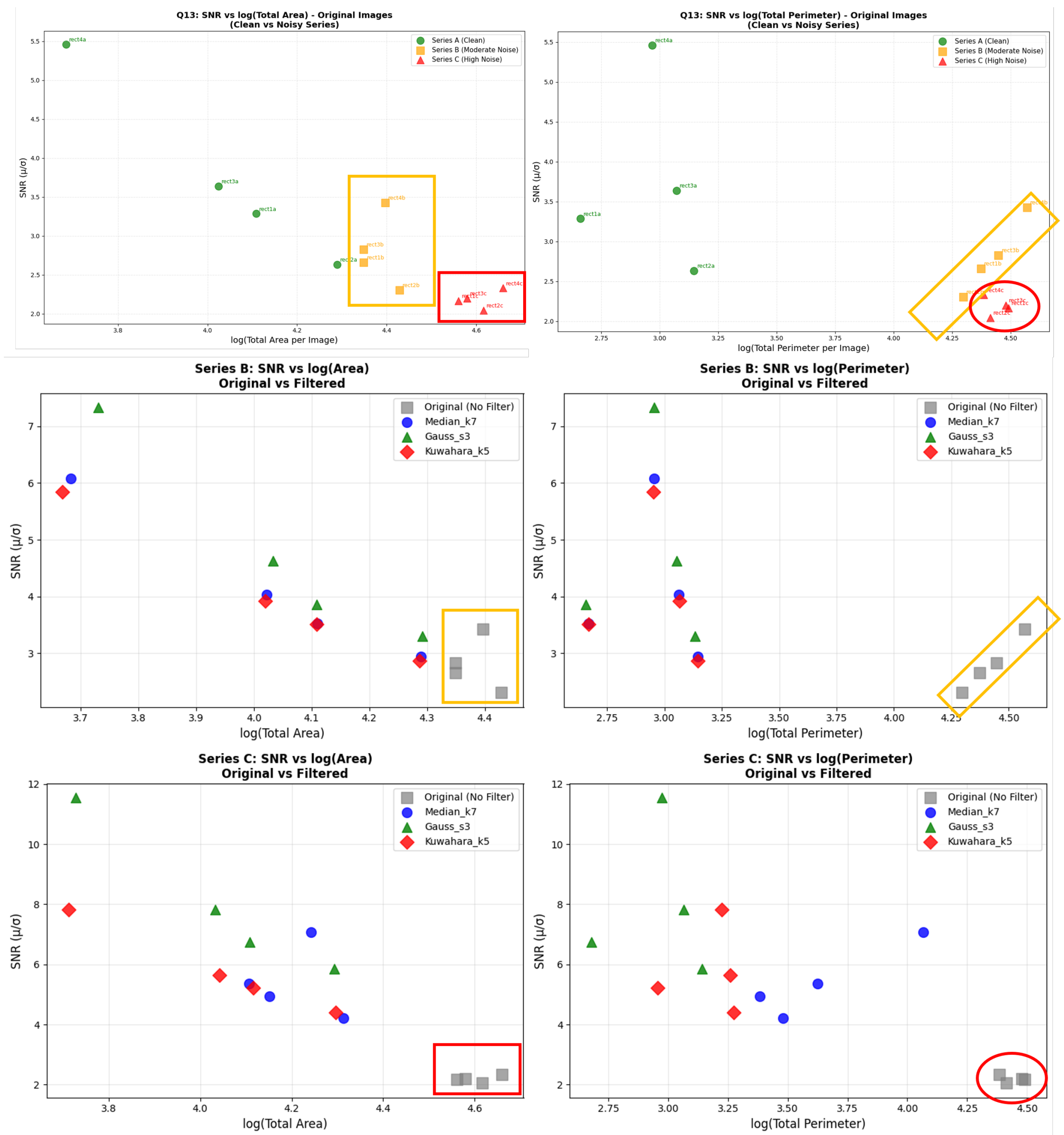

Figure 9: SNR versus log(total measured area) after filtering. Higher SNR correlates with more accurate area measurements (closer to ground truth).

Series B filtered (middle row). After filtering, the original (unfiltered) Series B points remain clustered in the orange box at high log(area) and low SNR; these are the same corrupted measurements as above. The filtered points (Median k7, Gauss s3, Kuwahara k5) shift towards lower log(area) values and higher SNR, meaning that filtering both improves signal quality and brings measurements closer to the ground truth. Gauss s3 (green triangles) consistently achieves the highest SNR among the filters shown, while Median k7 and Kuwahara k5 achieve intermediate improvement. The spread of points along a negative-slope trajectory reflects the object-size effect: smaller objects (rect4) achieve higher post-filter SNR but have lower total area.

Series C filtered (bottom row). The pattern is more dramatic for Series C. The unfiltered original points (grey squares, red box) cluster at very high log(area) and very low SNR (≈ 2.1–2.3), indicating heavily corrupted measurements. After filtering, Gauss s3 achieves the largest SNR improvement and simultaneously recovers measurements closest to the true values. However, even the best filter cannot fully recover small objects: rect4c points remain scattered, reflecting that the noise level in Series C exceeds the effective range of these filters for small objects. The perimeter plots (right column) show greater scatter than the area plots throughout, consistent with perimeter being more sensitive to boundary-level noise as established in Part 2.2.

Across all plots, a consistent pattern emerges: higher post-filter SNR correlates with log(measurement) values closer to the ground truth. Filtering moves points out of the corrupted cluster (orange/red boxes) towards the expected measurement range. The Gaussian filter with $\sigma = 3$ provides the best recovery in both series, while small-kernel and edge-preserving filters (Kuwahara k5, Median k7) provide partial but incomplete recovery, particularly for Series C.

References

[1] Claude E Shannon. Communication in the presence of noise. *Proceedings of the IRE*, 37(1):10–21, 2006.